

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

May 2017

EC203.01/EC203 Solid State Electronics

Date: 25.05.2017, Thursday

Time: 1:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) What is mobility? Derive equation for current density J and conductivity σ . [08]

Q - 2 (a) Draw the circuit diagrams of a forward-biased and a reverse-biased p-n junction diode. Draw the V-I characteristic curve and explain its nature. [06]

(b) How half wave rectifier demerits are overcome from center tap full wave rectifier? [07]
Draw and explain circuit diagram of full wave rectifier using centre tap transformer.
Derive its mathematical equation for maximum efficiency.

OR

Q - 2 (a) Define following. [01]
1. Maximum Forward Current [01]
2. Peak inverse voltage [01]
3. Maximum Power rating [01]
4. Forward Resistance [01]
5. Diffusion capacitance [01]
6. Ripple factor [01]

(b) Describe the filter action of the following filter circuits. [02]
1. Shunt capacitor filter [02]
2. choke input filter [02]
3. capacitor input filter [03]

Q - 3 (a) 1. The base of transistor is _____ doped. [01]
a) heavily b) moderately c) lightly d) none of the above
2. In an NPN transistor, _____ are the minority carriers. [01]
a) free electrons b) holes c) donor ions d) acceptor ions
3. The output impedance of transistor connected in _____ arrangement is the highest. [01]
a) common emitter b) common collector c) common base d) none of the above
4. The number of depletion layers in a transistor is _____. [01]
a) four b) three c) one d) two
5. A transistor has _____. [01]
a) one pn junction b) two pn junction c) three pn junction d) four pn

junction

6. The value of α of a transistor is _____. [01]
a) more than 1 b) less than 1 c) 1 d) none of the above
7. A transistor is _____ operated device. [01]
a) current b) voltage c) both voltage and current d) none of the above
- (b) How will you draw d.c. load line on the output characteristics of transistor? What is its importance? [07]

OR

- Q - 3 (a) Draw and explain input and output characteristics of common base (CB) transistor connection. What do you infer from these characteristics? [07]
- (b) How transistor works as an amplifier? Describe it with necessary diagram. [07]

SECTION – II

- Q - 4 (a) Define and Derive expression of R_i , R_o , A_i , A_v , and A_p for hybrid Common Emitter amplifier. [08]
- Q - 5 (a) Explain the construction and working of metal oxide semiconductor field effect transistor (MOSFET). [06]
- (b) Explain JFET parameter with its detail. Derive relationship between them. [07]

OR

- Q - 5 (a) Describe construction and working of junction field effect transistor with necessary diagram. [06]
- (b) In a self-bias n- channel JFET, the operating point is to be set at $I_D=1.5\text{mA}$ and $V_{DS}=10\text{V}$. The JFET parameters are $I_{DSS}=5\text{mA}$ and $V_{GS(off)}=-2\text{V}$. Find the values of R_S and R_D . Given that $V_{DD}=20\text{V}$. [07]
- Q - 6 (a) Explain the following terms: [02]
(1) Frequency response [02]
(2) Decibel gain [03]
(3) Bandwidth gain [03]
- (b) Draw and explain circuit diagram of transformer coupled multi stage amplifier. Enlist its advantages and disadvantages. [07]

OR

- Q - 6 (a) Describe following distortion of an amplifier [04]
1) frequency [03]
2) phase [04]
- (b) What do you understand by multi stage transistor amplifier? Mention its need. [04]
- Explain what will happen to the voltage gain of an amplifier if the bypass capacitor is open. [03]
